

Intro

For part 2 of my exploration of Japanese tourism volumes, I built an ETS forecasting model for Japan international visitor volume and evaluated its 24-month out-of-sample performance.

1. Exploring Exponential Smoothing Models

In the previous installment, I explored some aspects of past international visitor volume to Japan. Now I want to utilize that data to create specific predictions about visitor volume.

The statistical method I wanted to utilize is called exponential smoothing. It's a prediction method that uses a weighted average of past data points to make future predictions. The data points that come further from the past have progressively weaker weights. Exponential smoothing is popular in time-series prediction creation because it's simple to use and its emphasis on recent trends in the data has produced good empirical results.

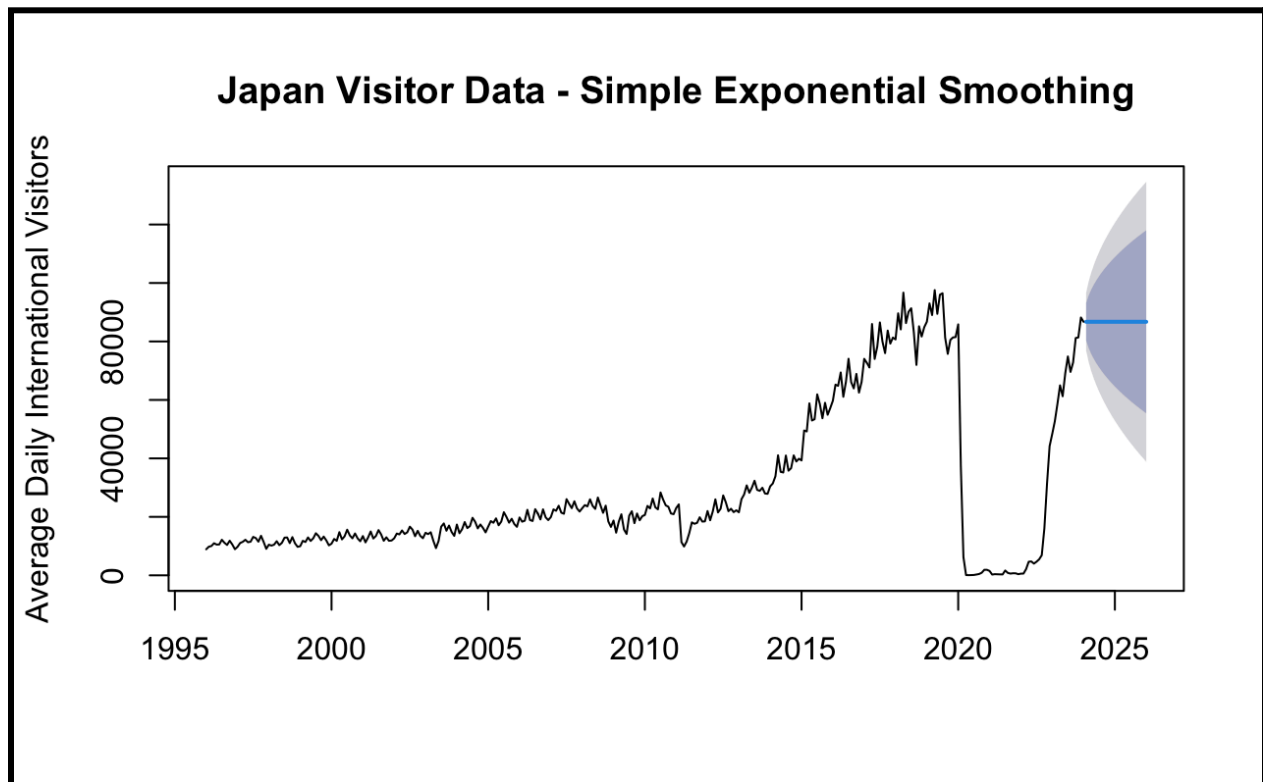
While I used Python in the previous section, I opted to use R here. This is because R contains the `forecast` package, which makes the creation and visualization of exponential smoothing models easy. I also wanted to test the efficacy of my model after model selection, so I set aside the two most recent cycles of data for testing (Feb 2024 - Jan 2026, inclusive).

Simple exponential smoothing is a technique that utilizes past data to create a “flat” point forecast for future time points. It assumes no significant trend or seasonal component to the data. The formula for arriving at the next successive predictive value for our simple exponential smoothing is as follows:

$$\hat{y}_{T+1|T} = \alpha y_T + (1 - \alpha) \hat{y}_{T|T-1}$$

Note: all equations such as the one above are taken from the free textbook [Forecasting: Principles and Practice](#) (see notes at the end)

The two parameters needed for simple exponential smoothing are the weight α and the initial value l_0 . These parameters can be optimized with the `forecast` package, which uses SSE (sum of squared errors) as a loss function. Applying simple exponential smoothing to the Japan dataset would seem inappropriate because the data displays a trend and seasonal variation, but it serves as a good visual demonstration of the method.



The point forecast, which begins with Feb. 2024, is 86,725.25 average daily international visitors, continuing in perpetuity. While the trend remains flat, the prediction interval increases over time, reflecting the increasing uncertainty that comes with making predictions further in the future.

Let's increase the complexity of the model. Based on the data that we have, it appears that international visitors levels are trending upwards, so let's add a trend into our model. Like with the level, we can model the trend at a certain time stamp using a weighted average of the previous trends.

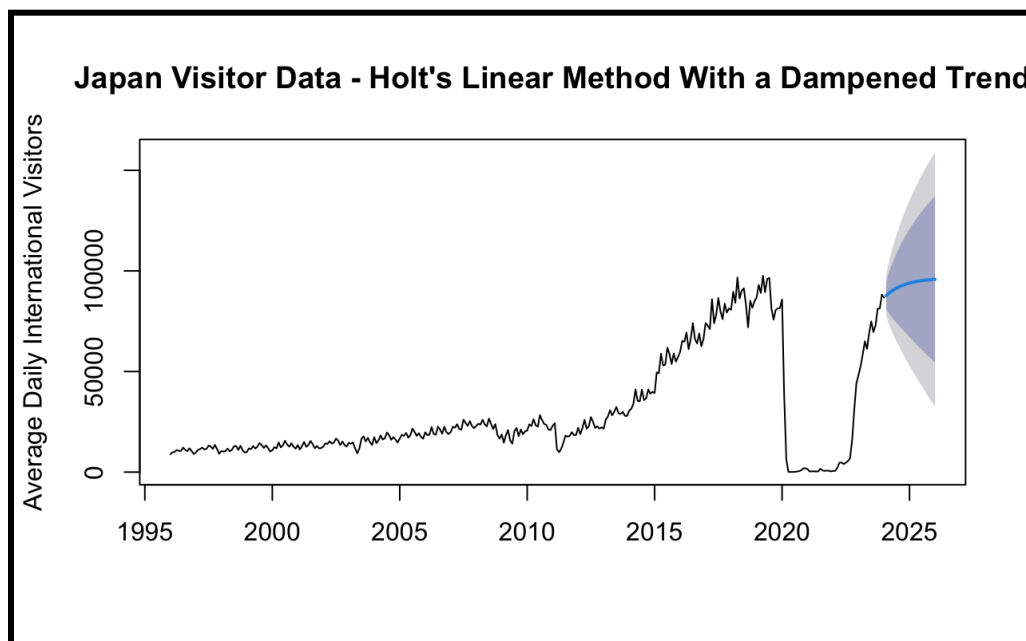
Forecast equation	$\hat{y}_{t+h t} = \ell_t + hb_t$
Level equation	$\ell_t = \alpha y_t + (1 - \alpha)(\ell_{t-1} + b_{t-1})$
Trend equation	$b_t = \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)b_{t-1},$

As before, we can use the `forecast` package to optimize the trend's weight β^* and initial state b_0 . This linear trend method is known as Holt's method. This method can be informative, but it also tends to "overforecast" future results. Therefore, many working with time series prefer

a “dampened” trend approach, where the effect of the trend is damped over time. This necessitates the introduction of another dampening parameter $0 < \phi < 1$.

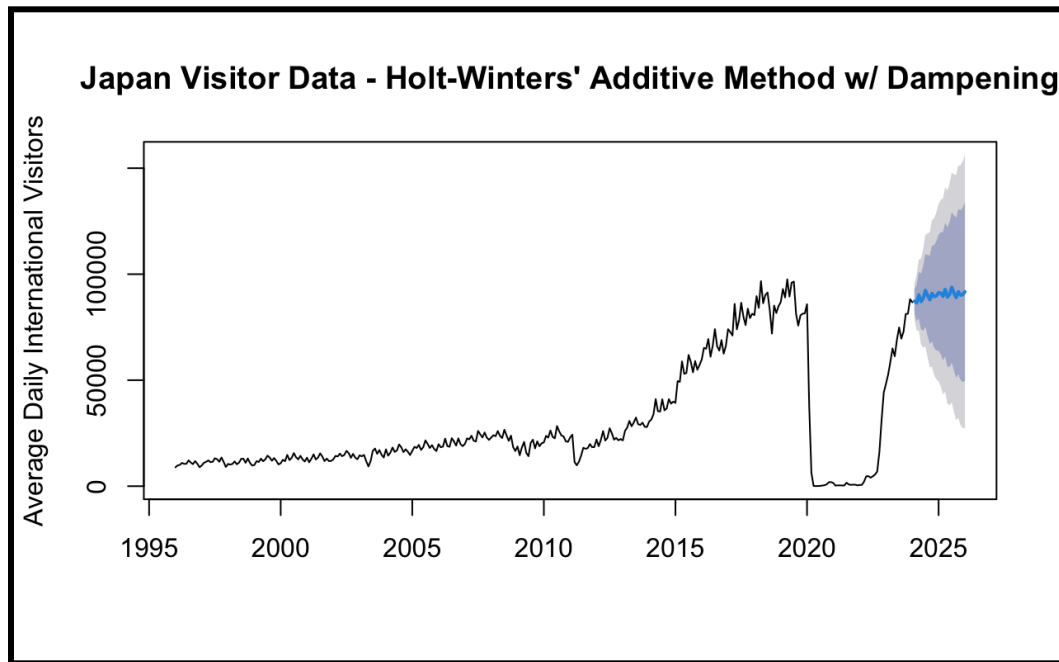
$$\begin{aligned}\hat{y}_{t+h|t} &= \ell_t + (\phi + \phi^2 + \dots + \phi^h)b_t \\ \ell_t &= \alpha y_t + (1 - \alpha)(\ell_{t-1} + \phi b_{t-1}) \\ b_t &= \beta^*(\ell_t - \ell_{t-1}) + (1 - \beta^*)\phi b_{t-1}.\end{aligned}$$

The forecast package also includes support for a dampened-trend prediction. Let’s try it out.



Going off the eye-test, this prediction seems more reasonable than the flat forecast we created earlier. The projected increase in tourism is quite slow. It's possible this caution is due to the precipitous drop in visitors during COVID.

One aspect that both methods lack, however, is an incorporation of the seasonal variation in the data. We can model the seasonal effect as yet another weighted average of past seasonal effects and introduce two more parameters: s as the initial seasonal effects, and γ as the seasonal weight. A question to ask when creating a prediction with a seasonal component is whether the trend is additive or multiplicative. Although the variability would appear to be multiplicative (the variations increase along with the level), I will model this with additive variability, because the near-zero visitation levels during COVID caused extremely large prediction errors when I modeled it with multiplicative errors.

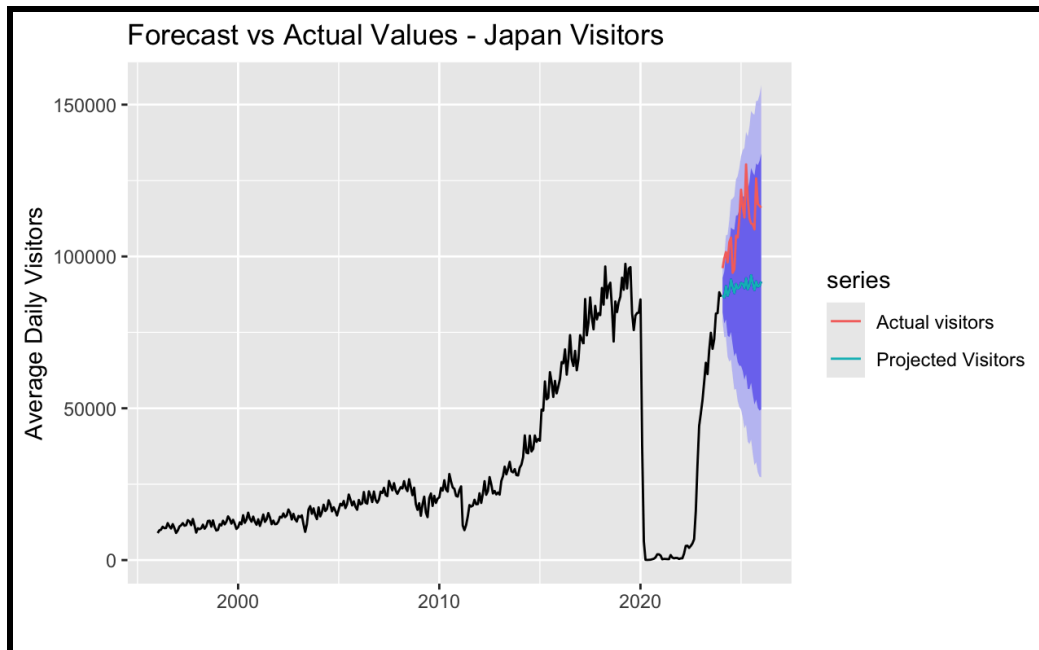


This prediction looks somewhat similar to the previous prediction, albeit this time with some seasonal variation.

2. Model Selection and Prediction

We need a way to select the best model based on the training data. Hyndman and Athanasopoulos recommend using corrected Akaike's Information Criterion (AICc) as an information criterion when selecting a model, which weights the sum of squared errors in concert with the number of parameters used.

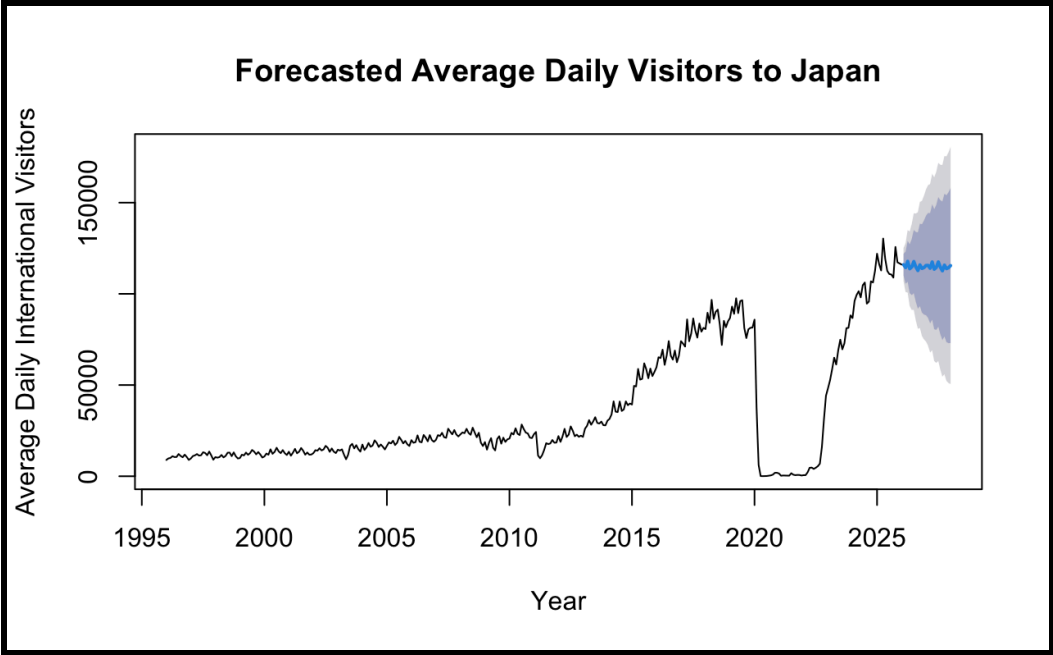
The `fpp3` package selects the best model according to the AICc. Using the `fpp3` package's optimization feature on the training data returns the Holt-Winters' Additive Method w/ Dampening model that we just created. This means it was the best model as judged by AICc. Let's compare the model's predictions with the actual visitor data.



Compared to the real-life data, our model underestimated the amount of visitors. The root mean squared error (RMSE) was 21,656 daily visitors amongst the 24 months that we predicted for. The mean absolute percentage error (MAPE) was 17.6%, meaning the model's predictions were off by that much on average. These are relatively large errors, and should be cause for some concern. However, given that the actual volume of visitors seemed to mostly lie within the 80% prediction interval of the model for the most part, and given that the model was asked to make predictions after a major disruption (COVID), I think these results are acceptable.

ME	RMSE	MAE	MPE	MAPE
19973.1	21656	19973.1	17.6	17.6

Let's use the model to create predictions of (currently) unreleased visitor volumes.



	Point Forecast	Lo 80	Hi 80
Feb 2026	115991.9	110065.3	121918.4
Mar 2026	114378.7	105615.5	123141.8
Apr 2026	117797.2	106637.5	128956.8
May 2026	113757.4	100422.1	127092.7
Jun 2026	114708.6	99341.9	130075.2
Jul 2026	117695.5	100406.1	134984.9
Aug 2026	114764.4	95641.2	133887.5
Sep 2026	112783.8	91903.4	133664.2
Oct 2026	115860	93290.5	138429.5
Nov 2026	113918.9	89722	138115.7
Dec 2026	114412.8	88645.3	140180.2
Jan 2027	115565.5	88279.9	142851.2
Feb 2027	115577.1	86822	144332.1
Mar 2027	114022.3	83843.7	144201
Apr 2027	117491	85931.6	149050.5
May 2027	113494.4	80594.3	146394.5
Jun 2027	114482.6	80279.7	148685.6

Jul 2027	117501.4	82031.1	152971.7
Aug 2027	114597.6	77893.5	151301.8
Sep 2027	112640.6	74734.1	150547
Oct 2027	115736.9	76658	154815.8
Nov 2027	113813.2	73590	154036.4
Dec 2027	114322	72981.1	155662.8
Jan 2028	115487.5	73054.2	157920.8

3. Conclusion

The model predicts a relatively flat trend in the number of future visitors to Japan. Going off the lowest point estimate, the least crowded time to visit Japan would be Sept 2027, with a projected average daily visitor count of 112,641. However, the intervals for that prediction are quite large, at [74,734, 150,547]. If you want a safer bet, you could try for Sept 2026, with a similarly low projected daily visitor count of 112,784, and a tighter margin prediction interval at [91,903, 133,664].

All told, however, there is not that much variation in the predicted visitation levels over the coming months. The difference between the projection for the most crowded month (April 2026) and the least crowded month (September 2027) is only 5,157 daily visitors.

This subverted my expectations going into this project. I was trained to think about Japanese tourism having busy and not-busy seasons. However, looking at both the recorded data and our predictions, it is clear that the effect of seasonal variation is miniscule when compared to the overall year-to-year trend (the COVID pandemic is a great demonstration of this fact). The flatness of the prediction is probably exacerbated by the fact that I had to model the data with additive seasonality, a consequence of the pandemic bringing visitations down to near-zero level.

In the future, it would be helpful to include variables other than past visitation levels. I suspect macro-economic trends both within and outside of Japan contribute significantly to visitation levels (for instance, the relative strength of the Japanese Yen). It would also be interesting to encode anomalous events, such as pandemic, or Japan's recent diplomatic row with China, into the model. A dynamic regression model could be a good way to achieve this.

Sources

- Equations and theory
 - Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice (3rd ed.)*. OTexts. <https://otexts.com/fpp3/>
- Japan Tourism data
 - Japan Tourism Marketing Co. (2026). Statistics of Overseas Residents' Visits to Japan (Historical figures/by country/by purpose) [Data set]. https://www.tourism.jp/wp/wp-content/uploads/2026/03/JTM_inbound_20260304_eng.xlsx